CHIDU UMEH

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Professional Summary: Physics M.S. candidate with hands-on experience building predictive models, BI dashboards, and GenAI-powered tools across financial and scientific domains. Proficient in Python and SQL with a track record of extracting actionable insights from large, complex datasets and communicating findings clearly to technical and non-technical audiences.

SKILLS

Tools: LaTeX, Tableau, Microsoft Office (Excel, PowerPoint), Jupyter Notebooks, Creo CAD
Specialties: Machine Learning, Statistical Modeling, Generative AI, Prompt Engineering, Data Visualization, BI Development, SQL Querying, Cross-Functional Communication
Languages: Python (Pandas, NumPy, Scikit-learn), SQL (PostgreSQL), Git, MATLAB, R
Libraries: NumPy, Pandas, Matplotlib, TensorFlow

PROFESSIONAL EXPERIENCE

Erdos Institute, Quant Finance Bootcamp — **September 2025 – November 2025**

- Designed and built a Python financial modeling tool from scratch applying the Black-Scholes-Merton framework to compare market-implied vs. realized volatility for VanEck Africa ETF (AFK), delivering working code and a written walkthrough as tangible artifacts.
- Quantified a volatility risk premium (IV > RV, correlation 0.27) through delta hedging simulations and delivered structured visualizations to a panel of 52 team evaluators. Awarded Top Project out of 52 submissions.

University of Central Florida — **January 2025 – Present**

Graduate Teaching Assistant

- Delivered weekly instruction and lab support for 250+ students, explaining technical concepts clearly in classroom and laboratory settings.
- Guided students through analytical problem solving, interpretation of results, and structured technical communication.

Graduate Research Assistant

- Designed a dual-axis rotation mount in Creo CAD to enable simultaneous angular control of a microwave source and signal detector, expanding experimental capability in an active spintronics research lab.

U.S. Gun Violence Intelligence Tool — **Spring 2026**

- Deployed an interactive geospatial analytics platform processing 300,000+ mortality records (1999-2024), applying race and sex stratified statistical breakdowns across all 50 states to surface public health patterns in a large heterogeneous dataset.
- Built end-to-end in Python using LLM-based coding assistants (Claude, ChatGPT Codex), integrating data ingestion, cleaning, statistical modeling, and an interactive front-end visualization layer accessible to non-technical audiences.

Autura — **Summer 2023**

Data Analyst Internship

- Used SQL to extract, clean, and normalize auction and logistics datasets across multiple sources, ensuring data integrity for downstream analysis and leadership reporting.
- Built Tableau dashboards integrating U.S. Census data with internal records, surfacing regional cost patterns and logistics trends that informed leadership's market expansion decisions.

Bates College — **Mar 2019 – May 2022**

Research Data Analyst

- Modeled galaxy spectra in Python and developed reproducible pipelines to estimate stellar mass, star-formation rates, and dust content across tens of thousands of observations, enabling multi-institutional research.
- Delivered technical outputs through a thesis, conference presentation, and peer reviewed publication.

EDUCATION

University of Central Florida, Orlando, FL — **2026**
Master of Science in Physics (Concentration: Spintronics)
Bates College, Lewiston, ME — **2022**
Bachelor of Science in Physics (Minors: Computer Science & Applied Mathematics)